GM UNIVERSITY
(Established under the Karnataka State Act No. 19 of 2023)
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Faculty of Engineering and Technology

School of Computer Science and Technology

Module 5
Chapter 3: Artificial Neural Networks (ANN)

Introduction

Artificial Neural Networks (ANNs) are computational models inspired by the structure and functioning of the human brain.They consist of interconnected processing units called neurons that learn patterns from data.ANNs are capable of classification, prediction, pattern recognition, speech processing, image analysis, medical diagnosis, recommendation systems, and forecasting.

Characteristics:
• Learning from examples
• Adaptive behavior
• Parallel processing
• Fault tolerance
• Generalization capability

Biological Neurons
A biological neuron is the fundamental processing unit of the nervous system.

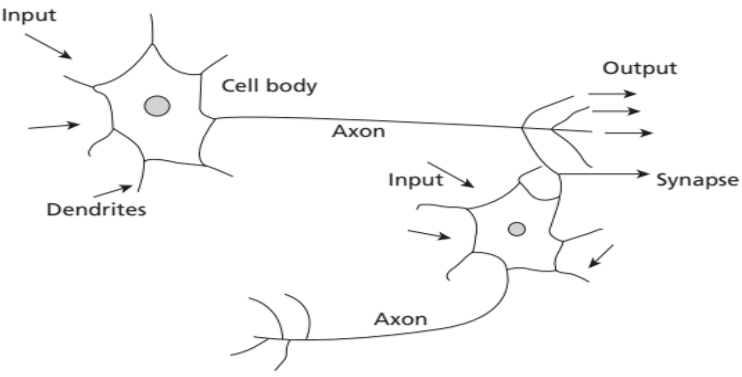


Figure 10.1: A Biological Neuron

Functions:

• Dendrites receive signals.

• Soma integrates information.

• Axon transmits impulses.

• Synapses connect neurons.

The brain contains billions of interconnected neurons forming a highly complex processing network.

Artificial Neuron Model

Artificial neurons mimic the behavior of biological neurons.

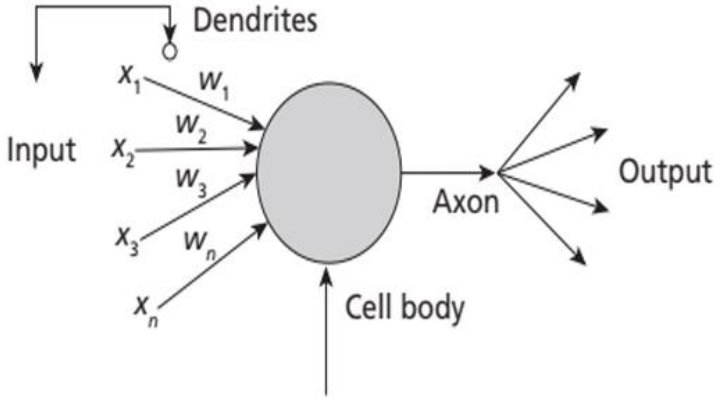


Figure 10.2: An Artificial Neuron

Operation:

Net Input = $\Sigma(w_i x_i) + b$

where:

xi = inputs
wi = weights
b = bias

The activation function transforms the weighted sum into an output signal.

ANN Structure
An ANN is organized into layers.

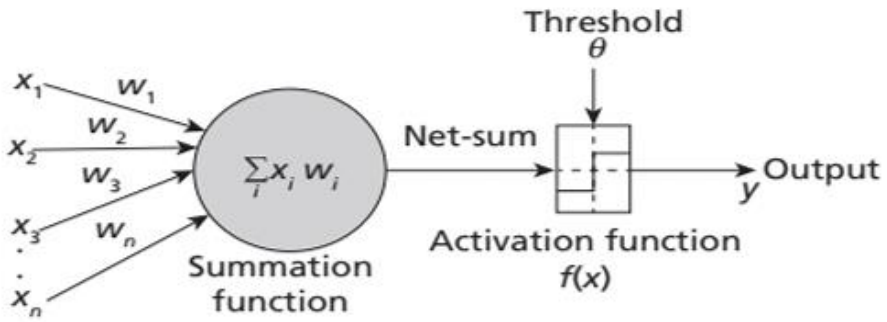


Figure 10.3: McCulloch & Pitts Neuron Mathematical Model

Components of a Perceptron

The Perceptron is the simplest form of an Artificial Neural Network (ANN). It consists of four major components:

1. Inputs
2. Weights and Bias
3. Net Sum (Summation Function)
4. Activation Function

1. **Inputs**: are the signals or data values received by a neuron from the external environment or from other neurons. They represent the features or attributes of a problem.

   Explanation: Inputs are denoted by x_1, x_2, x_3, ..., x_n. Each input corresponds to a particular characteristic of the data being processed.

   The input layer of a neural network receives these values and forwards them to the neuron for further processing.

Importance

- Inputs provide information to the neuron.
- The quality of inputs directly affects the performance of the neural network.
- More relevant inputs generally lead to better learning and prediction.

2. Weights and Bias:

Weights are numerical values assigned to inputs that indicate their importance in the decision-making process. Bias is an additional parameter used to shift the activation function and improve learning.

Each input is associated with a weight.

$w_1, w_2, w_3, \ldots, w_n$

A weight determines how strongly an input influences the output.

Bias

Bias is an additional value added to the weighted sum before applying the activation function.

Purpose of Bias:

- Helps shift the decision boundary.
- Allows the neuron to activate even when all inputs are zero.
- Improves flexibility and learning capability.

Importance

- Weights determine the importance of each feature.
- Bias improves the accuracy of classification.
- Learning in neural networks mainly involves adjusting weights and bias values.

3. Net Sum (Summation Function)

Definition

The Net Sum is the weighted combination of all inputs received by a neuron. It is calculated by multiplying each input by its corresponding weight and then adding the results together along with the bias.

Mathematical Expression

Net=∑i=1nwixi+bNet=\sum_{i=1}^{n} w_i x_i+bNet=∑i=1nwixi+b

Explanation

The summation unit performs the following operations:

1. Multiply each input by its corresponding weight.
2. Add all weighted inputs.
3. Add the bias value.

The resulting value is called the Net Input or Net Sum.

4. Activation Function

Definition

An Activation Function is a mathematical function that determines whether a neuron should produce an output and what that output should be.

Explanation

After computing the Net Sum, the neuron must decide whether it should be activated (fire) or not. This decision is made by the activation function.

The activation function transforms the net input into an output value.

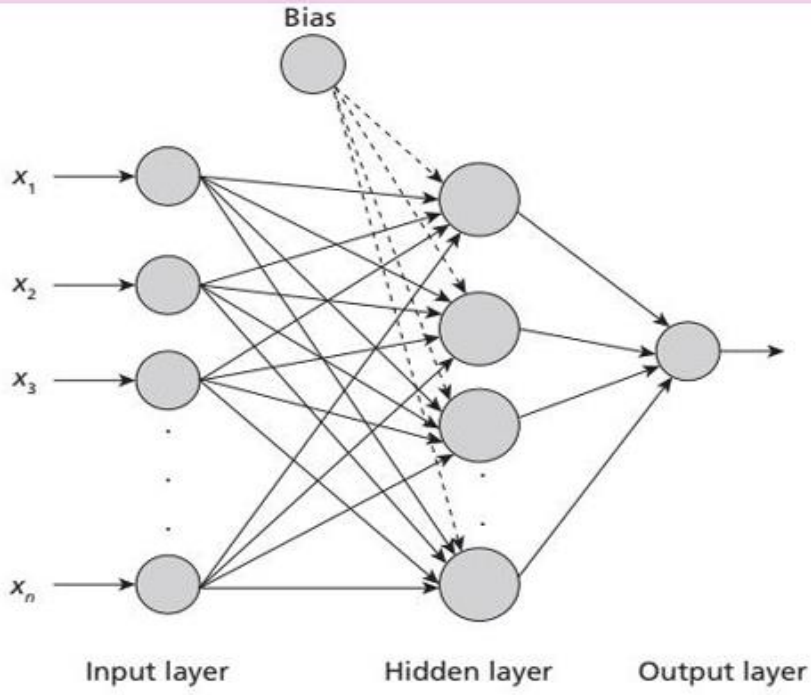


Figure 10.4: Artificial Neural Network Structure

Activation Functions

Activation functions determine the output of a neuron and introduce non-linearity into the network.

Why Activation Functions are Needed?

Without activation functions, a neural network behaves like a simple linear model regardless of the number of layers.

Activation functions allow neural networks to learn complex and non-linear relationships.

1. Identity Function or Linear Function

$$f(x) = x \ \forall x \tag{10.4}$$

The value of $f(x)$ increases linearly or proportionally with the value of x. This function is useful when we do not want to apply any threshold. The output would be just the weighted sum of input values. The output value ranges between $-\infty$ and $+\infty$.

2. Binary Step Function

$$f(x) = \begin{cases} 1 & if\ f(x) \geq \theta \\ 0 & if\ f(x) < \theta \end{cases} \tag{10.5}$$

The output value is binary, i.e., 0 or 1 based on the threshold value θ. If value of $f(x)$ is greater than or equal to θ, it outputs 1 or else it outputs 0.

3. Bipolar Step Function

$$f(x) = \begin{cases} 1 & if\ f(x) \geq \theta \\ -1 & if\ f(x) < \theta \end{cases} \tag{10.6}$$

The output value is bipolar, i.e., +1 or −1 based on the threshold value θ. If value of $f(x)$ is greater than or equal to θ, it outputs +1 or else it outputs −1.

4. Sigmoidal Function or Logistic Function

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{10.7}$$

It is a widely used non-linear activation function which produces an S-shaped curve and the output values are in the range of 0 and 1. It has a vanishing gradient problem, i.e., no change in the prediction for very low input values and very high input values.

5. Bipolar Sigmoid Function

$$\sigma(x) = \frac{1 - e^{-x}}{1 + e^{-x}} \tag{10.8}$$

It outputs values between −1 and +1.

6. Ramp Functions

$$f(x) = \begin{cases} 1 & if\ x > 1 \\ x & if\ 0 \leq x \leq 1 \\ 0 & if\ x < 0 \end{cases} \tag{10.9}$$

It is a linear function whose upper and lower limits are fixed.

Perceptron and Learning Theory

The perceptron is the earliest neural network model proposed by Frank Rosenblatt.

Architecture:

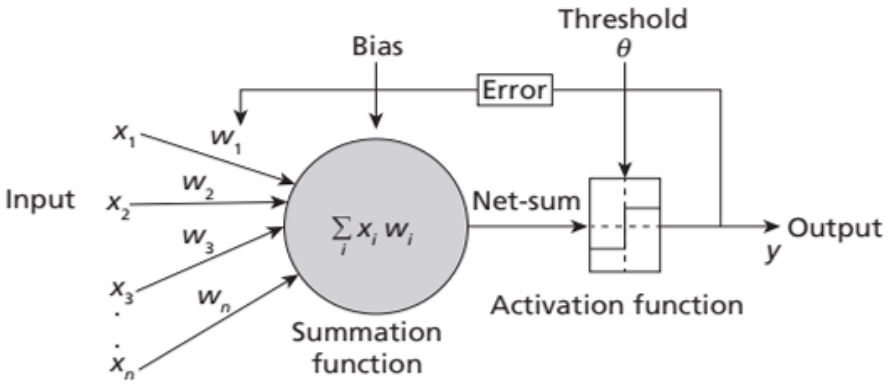


Figure 10.5: Perceptron Model

Characteristics:
• Single-layer network
• Supervised learning
• Binary classification

Learning Theory:
The perceptron learns by adjusting weights to reduce prediction errors.
The objective is to correctly classify training examples.

Limitations:
• Can only solve linearly separable problems.
• Cannot solve XOR problems.

Perceptron Algorithm

Algorithm 10.1: Perceptron Algorithm

Set initial weights $w_1, w_2, \ldots, w_n$ and bias θ to a random value in the range [–0.5, 0.5].

For each Epoch,

1. Compute the weighted sum by multiplying the inputs with the weights and add the products.
2. Apply the activation function on the weighted sum:
$$Y = \text{Step}((x_1 w_1 + x_2 w_2) - \theta)$$
3. If the sum is above the threshold value, output the value as positive else output the value as negative.
4. Calculate the error by subtracting the estimated output $Y_{estimated}$ from the desired output $Y_{desired}$:
$$\text{error } e(t) = Y_{desired} - Y_{estimated}$$
[If error $e(t)$ is positive, increase the perceptron output Y and if it is negative, decrease the perceptron output Y.]
5. Update the weights if there is an error:
$$\Delta w_i = \propto \times e(t) \times x_i$$
$$w_i = w_i + \Delta w_i$$
where, x_i is the input value, $e(t)$ is the error at step t, $\propto$ is the learning rate and Δw_i is the difference in weight that has to be added to w_i.

Advantages:

• Simple implementation.

• Fast convergence for linearly separable data.

Disadvantages:

• Limited learning capability.

Types of Artificial Neural Networks

Feed Forward Neural Network (FFNN) : A Feed Forward Neural Network is the simplest type of neural network in which information flows only in one direction, from the input layer to the output layer through one or more hidden layers. There are no feedback connections or loops.

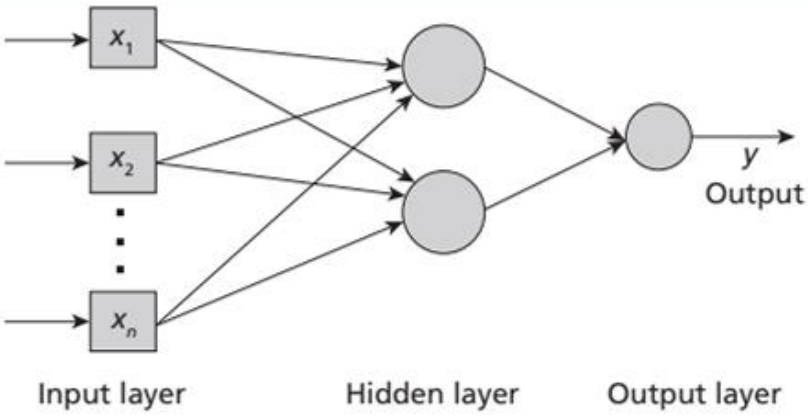


Figure 10.7: Model of a Feed Forward Neural Network

Characteristics

- **One-way flow of information.**
- **No memory of previous inputs.**
- **Simple architecture.**
- **Easy to train and implement.**

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2. Fully Connected Neural Network (FCNN)

In a Fully Connected Neural Network, every neuron in one layer is connected to every neuron in the next layer. This allows maximum information transfer between layers.

Characteristics

- Dense connections between layers.
- Every neuron participates in learning.
- Large number of trainable parameters.

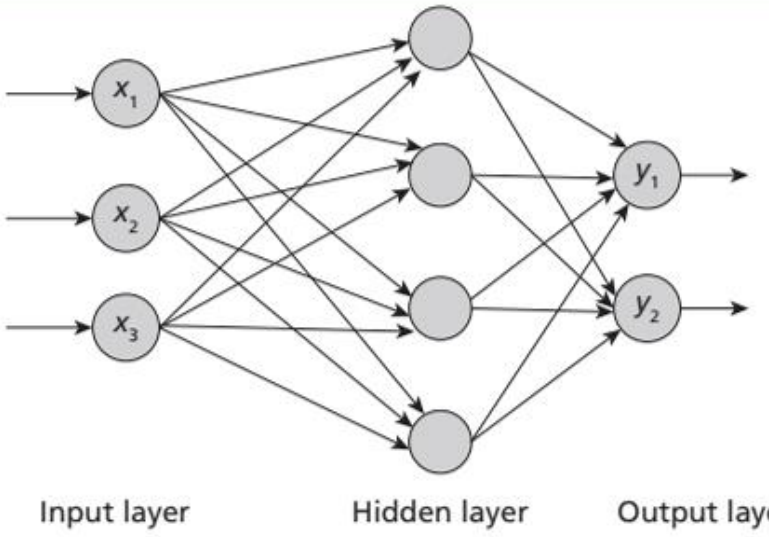


Figure 10.8: Model of a Fully Connected Neural Network

3. Multi-Layer Perceptron (MLP)

A Multi-Layer Perceptron is an extension of the perceptron that contains one or more hidden layers between the input and output layers. It uses non-linear activation functions and is trained using the Backpropagation algorithm.

Characteristics

- Contains multiple hidden layers.
- Supports supervised learning.
- Can solve non-linearly separable problems.
- Uses activation functions such as Sigmoid, Tanh, or ReLU.

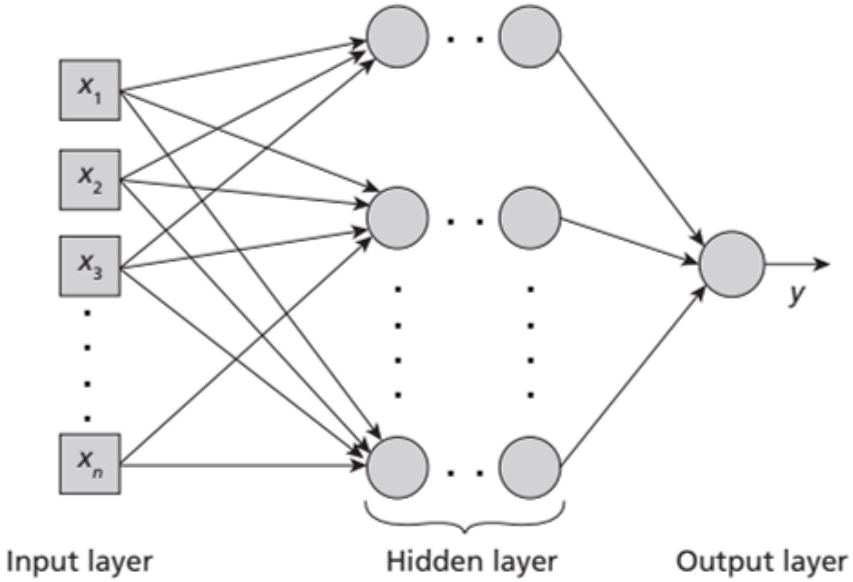


Figure 10.9: Model of a Multi-Layer Perceptron

4. Feedback Neural Network (Recurrent Neural Network)

A Feedback Neural Network contains feedback connections where the output of a neuron can be fed back into the network as input. This enables the network to remember previous information and process sequential data.

Characteristics

- Contains loops or feedback paths.
- Has memory of previous inputs.
- Information flows in both forward and backward directions.
- Suitable for temporal and sequential data.

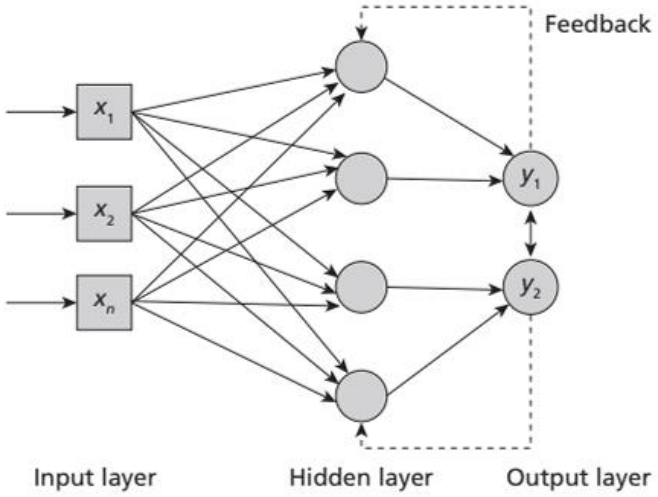


Figure 10.10: Model of a Feedback Neural Network